Minho Kim

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Scholar | ResearchGate | Twitter | ORCID

Research Interests

- **Geospatial Analysis:** Remote Sensing, Computer Vision, GIScience, Network Science
- **Machine Learning:** Deep Learning, GeoAI, Explainable AI
- **Environmental Planning:** Risk & Resilience, Natural Hazards, Sustainable Development

Education

University of California, Berkeley — **Sep 2021 – Present**
Ph.D. Environmental Planning
Dissertation: Data-Driven Planning for Resilience Against Natural Hazard Risks
Advisors: Marta C. González, John Radke
Exam Committee: John Radke, Marta C. González, Iryna Dronova, Solomon Hsiang

Seoul National University — **Mar 2017 – Feb 2021**
M.S. Civil & Environmental Engineering
Thesis: Local Climate Zone Classification Using Multi-Scale Convolutional Networks
Advisor: Yongil Kim

Seoul National University — **Sep 2012 – Feb 2017**
B.S. Civil & Environmental Engineering
Thesis: Monitoring North Korea's 4th Nuclear Test Site with Sentinel-1A Data Using DInSAR
Advisor: Yongil Kim

Research Experience

Researcher (Center for Catastrophic Risk Management) — **Jan 2025 – Present**
- System-level governance frameworks for catastrophic risk management
- Reviewed natural hazard modeling and data sources for catastrophic wildfire events

Visiting Researcher (Catalan Fire Service) — **July 2024 – Sep 2024**
Advisor: Marc Castellnou — *Barcelona, Spain*
- Modeled fire potential polygons and networks using simulations and hydrology-inspired tools for fire suppression decision support [W2]
- Improved existing WUI maps using automatic structural separation distance and permeability metrics
- Measured shared responsibility metrics in WUI map areas in Catalonia [W3]

Graduate Student Researcher (River Lab, Funded by CalTrans) — **May 2023 – June 2024**
Advisors: Mathias Kondolf, John Radke — *UC Berkeley*
- Developed methodology to estimate bulking factors and protect critical infrastructure against debris flows [W5]
- Built GUI used to aggregate data APIs and GIS data layers related to post-fire debris flow probability

Graduate Student Researcher (HuMNet Lab, Funded by C3.AI) — **Jan 2022 – present**
Advisor: Marta C. González, Mentor: Cristobal Pais — *UC Berkeley*
- Generated physics-based, semi-empirical data computed in *R* to integrate into a cellular automata simulator (*C++*) to conduct fire spread simulations [P6].
- High resolution mapping of fuels and vegetation using deep learning [C10].

Research Assistant (SPINS-RS Lab) — **Mar 2019 – Feb 2021**
Advisor: Yongil Kim — *Seoul National University*
- Urban Remote Sensing: Generated high resolution Local Climate Zone classification maps multi-scale CNNs (∼80% accuracy) [C7] and multi-scale, multi-level attention CNNs (∼90% accuracy) trained with multitemporal Sentinel-2 images and multi-modal GIS data [P4].
- Renewable Energy: Predicted photovoltaic power of solar farms with high precision ($<$ 5% Normalized MAE) using large-scale time series of multitemporal geostationary satellite images and multi-source meteorological data [C2], [C4], [P3].

- Image Fusion: Developed a spatiotemporal image fusion model in Matlab to produce disaggregated Landsat-8 thermal images in heterogeneous urban areas [C6].
- Change Detection/Monitoring: Applied radiometric calibration to help detect and monitor burn scars using change detection results from multitemporal Sentinel-2 and PlanetScope images [C3], [P2].

Undergraduate Research Assistant (SPINS-RS Lab) — **Aug 2016 – Feb 2017**
Advisor: Yongil Kim — *Seoul National University*

- Analyzed ground deformations in inaccessible, remote areas using dInSAR with Sentinel-1 SAR images.
- Carried out fieldwork and experiments using a ground-based hyperspectral imager to monitor crop health.

Research Assistant (Lawson Health Research Institute) — **Sep 2011 – Jan 2012**
Advisor: Jeffrey Carson — *London, Canada*

- Photoacoustic image reconstruction of a line source using multiple regularization percentages with maximum intensity projection using Matlab.

Work Experience

Researcher at Institute of Construction & Env. Eng. — **Mar 2021 – Aug 2021**
Advisor: Yongil Kim — *Seoul National University*

- Developed high resolution land cover maps of inaccessible areas using deep learning with very high resolution satellite imagery [C8].

PR Manager — **Mar 2021 – Aug 2021**
Education & Research Program (InfraSPHERE) — *Seoul National University*

Lab Manager — **Mar 2021 – Aug 2021**
SPINS-RS Lab — *Seoul National University*

Honors & Awards

Outstanding Graduate Student Instructor Award — **April 2024**
UC Berkeley (GSI Teaching and Resource Center)

ICE-KSCE Master's Thesis Award — **July 2021**
Institution of Civil Engineers (UK) & Korean Society of Civil Engineers

Best Student Paper Award at ISRS2021 — **May 2021**
Korean Society of Remote Sensing and Gaia3D

Environmental Geospatial Data Idea Contest (Excellence Award) — **Nov 2020**
Ministry of Environment, South Korea

SPINS Lab (Outstanding Research Award) — **Mar 2020**
Seoul National University

Student Competition using Meteorological Satellites (Research Award) — **Jan 2019**
Korean Meteorological Administration

Scholarships

AEP Student Scholarship — **July 2025**
Association of Environmental Professionals (AEP)

Beatrix C. Farrand Memorial Fellowship for Research — **May 2024**
UC Berkeley (Dept. of Landscape Architecture & Environmental Planning)

Beatrix C. Farrand Memorial Fellowship for Conference Travel — **May 2023**
UC Berkeley (Dept. of Landscape Architecture & Environmental Planning)

Robert N. Colwell Memorial Fellowship — **Feb 2023**
The American Society for Photogrammetry and Remote Sensing

Brain Korea 21 Plus Scholarship — **2019 – 2021**
National Research Foundation of Korea

Merit-based Scholarship — **2014–2017, 2019**
Seoul National University

National Scholarship for Science and Engineering — **2013 – 2014**
Korea Student Aid Foundation

SNU Global Scholarship — **2012 – 2013**
Seoul National University

Publications

* Equal contribution, † Corresponding author

Preprints & Working Papers

[W4] **Minho Kim**, Adrienne Dodd, John Radke, G. Mathias Kondolf. "Multi-criteria decision-making for cascading hazards: Case study of post-fire debris flows in Northern California".

[W3] **Minho Kim**, Tomàs Artés, Laia Estivill, Pau Guarque, Marta C. González, Marc Castellnou. "Next-generation wildfire risk management using deep learning embeddings and similarity search"

[W2] **Minho Kim**, Harrison Raine, John Radke, Marta C. González. "Rethinking Defensible Space: Spatial Responsibility for Wildfire Risk Mitigation in the Wildland Urban Interface. (To be submitted to *Landscape and Urban Planning*")

[W1] Cristobal Pais, **Minho Kim**, Yanyan Xu, John Radke, Marta C. González. "An interdisciplinary data-science approach to managing natural hazards risk. arXiv:2407.07270."

Peer Reviewed Journal Papers

[P7] **Minho Kim**, Marc Castellnou, Marta C. González. (2025). "Modeling potential fire spread polygons and networks for suppression strategies", *International Journal of Disaster Risk Reduction*.

[P6] **Minho Kim**, Cristobal Pais, Marta C. González. (2025). "Fire Spread Simulations Using Cell2Fire on Synthetic and Real Landscapes", *Scientific Reports*, 15 25173 (2025).

[P5] Xihan Yao, **Minho Kim**, Iryna Dronova, Joe McBride, G. Mathias Kondolf, John Radke. (2025). "Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification", *Landscape and Urban Planning*, 263, 105420.

[P4] **Minho Kim**, Jeong, D. & Kim, Y. (2021). "Local climate zone classification using a multi-scale, multi-level attention network", *ISPRS Journal of Photogrammetry and Remote Sensing*, 181, 345-366.

[P3] **Minho Kim**, Song, H. & Kim, Y. (2020). "Direct short-term forecast of photovoltaic power through a comparative study between COMS and Himawari-8 meteorological satellite images in a deep neural network", *Remote Sensing*, 12(15), 2357.

[P2] **Minho Kim**, Jung, M. & Kim, Y. (2019). "Histogram matching of Sentinel-2 spectral information to enhance Planetscope imagery for effective wildfire damage assessment", *Korean Journal of Remote Sensing*, 35(4), 517-534.

[P1] Kim, Y., **Minho Kim**, Choi, J. & Kim, Y. (2017). "Image fusion of spectrally nonoverlapping imagery using SPCA and MTF-based filters", *IEEE Geoscience and Remote Sensing Letters*, 14(12), 2295-2299.

Conference & Workshop Papers

[C16] **Minho Kim**, Harrison Raine, John Radke & Gonzalez, M.C. (2025). "A Network Modeling Approach to Quantify Homeowner Responsibility for Wildfire Risk Mitigation in the Wildland Urban Interface", *AGU 2025. New Orleans, USA. Dec 15-19, 2025.*

[C15] **Minho Kim**, Dodd, A., Radke J., Ulyashin V., Serra-Llobet, A., Falzone, A. & Kondolf, M. (2025). "Flood after Fires: A Scalable Decision-Support Framework for Post-Fire Debris Flow Risk", *AGU 2025. New Orleans, USA. Dec 15-19, 2025.*

[C14] **Minho Kim**, Pais, C., & Gonzalez, M.C. (2025). "Fire spread simulations using Cell2Fire on synthetic and real landscapes", *International Fire Ecology and Management Congress. Association for Fire Ecology. New Orleans, USA. Dec 2-6, 2025.*

[C13] Rut Domènech, **Minho Kim**, Pau Guarque, Laia Estivill & Marc Castellnou (2025). "Identifying optimal windows for prescribed burns using a deep learning model", *International Fire Ecology and Management Congress. Association for Fire Ecology. New Orleans, USA. Dec 2-6, 2025.*

[C12] **Minho Kim**, Castellnou, M. & Gonzalez, M.C. (2025). "Modeling fire potential networks for suppression strategies", *Conference of Complexity Sciences 2025 (Complex Systems, Data Analytics, and Human Behavior). Siena, Italy. Sep 2-5, 2025.*

[C11] Yao, X. & **Minho Kim**. (2023). "A Lidar-based Method for 3D Urban Forest Evaluation and Microclimate Assessment, a Case Study in Portland, Oregon, USA", *AGU23. American Geophysical Union. Dec 11-25, 2023.*

[C10] **Minho Kim**, Dronova, I. & Radke, J. (2023). "Semantic Segmentation of Enhanced Landform Maps Using High Resolution Satellite Images", *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023.* (*Attended as **Session Chair**)

[C9] Yao, X. & **Minho Kim** (2023). "Exploratory remote sensing data analysis and clustering of urban vegetation and land surface temperature in Portland, Oregon", *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023.*

[C8] **Minho Kim**, Kwak, T., Jung, J. & Kim, Y. (2021). "Mapping inaccessible areas using deep learning based semantic segmentation of VHR satellite images with OpenStreetMap data", *In Proceedings of the 2021 International Symposium of Remote Sensing, Virtual, May 26-28, 2021.*(*Awarded **Best Student Paper**)

[C7] **Minho Kim**, Jeong, D., Choi, H. & Kim, Y. (2020). "Developing High Quality Training Samples for Deep Learning Based Local Climate Zone Classification in Korea", *arXiv preprint, Presented at **AI for Earth Sciences Workshop at NeurIPS 2020**, Virtual, arXiv:2011.01436.*

[C6] **Minho Kim**, Cho, K., Kim, H. & Kim, Y. (2020). "Fusion of High Resolution Land Surface Temperature Using Thermal Sharpened Images from Regression-based Urban Indices", *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 3, (pp247-254).*

[C5] Song, A., Kim, C., **Minho Kim** & Kim, Y. (2019). "Analysis of Geospatial Technology for Smart City Development: Case Study of South Korea", *In Proceedings of The 1st Tunisian Smart Cities Symposium, Tunisia, 2019.*

[C4] Kim, G., Song, H., Kim, **Minho Kim** & Kim, Y. (2019). "Multimodal Merging of Satellite Imagery with Meteorological and Power Plant Data in Deep Convolutional Neural Network for Short-Term Solar Energy Prediction", *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*

[C3] **Minho Kim** & Kim, Y. (2019). "Integration of Sentinel-2 Spectral Information with High Spatial Resolution Planetscope Imagery for Wildfire Damage Assessment", *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*

[C2] Song, H., Kim, G., **Minho Kim** & Kim, Y. (2019). "Short-Term Forecasting of Photovoltaic Power Integrating Multi-Temporal Meteorological Satellite Imagery in Deep Neural Network", *In 2019 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Macao, (pp1-5).*

[C1] **Minho Kim** & Kim, Y. (2019). "Monitoring the Catastrophic 2018 Mendocino Complex Wildfire Using the Sentinel Constellation", *In Proceedings of the 2019 International Symposium of Remote Sensing, Taiwan, April 14-17, 2019.*

Invited Talks & Panels

[T6] **"Shared responsibility and structure separation distance using network modeling and metrics"**
Invited seminar for the Catalan Fire Service
Forest Actions Reinforcement Group (GRAF), Catalan Government **July 2024**
Invited seminar for the Pau Costa Foundation
Pau Costa Foundation **July 2024**

[T5] **"Fire spread modeling and superpixel-based fire suppression networks"**
Invited seminar for the Meteorology and Air Quality group
Wageningen University & Research **July 2024**
Invited seminar for the Catalan Fire Service
Forest Actions Reinforcement Group (GRAF), Catalan Government **July 2024**

[T4] **"Data-driven planning and modeling for wildfire research using geospatial data science and network science"**
Invited seminar for the Ecological Sensing Lab
Seoul National University **Aug 2025**
Guest lecture for CP4190 Introduction to Climate Change Planning
Georgia Institute of Technology **Feb 2024**

[T3] **"Exploring Research in the Environmental Field"**
Berkeley Environmental Economics and Policy Students
UC Berkeley **Oct 2021**

[T2] **"Urban Remote Sensing"**
Guest lecture for graduate course 457.544: Satellite Image Interpretation
Seoul National University **Apr 2020**

[T1] **"Urban Remote Sensing"**
Seminar for the Interdisciplinary Program in Landscape Architecture
Seoul National University **Jan 2020**

Patents & Software

Song, H., Kim, Y., **Minho Kim**, Kim, K. *Convolutional neural networks for short-term photovoltaic forecast using satellite imagery, meteorological data, and power station data.* Patent, South Korea, 2021.

Teaching

UC Berkeley

Lead Instructor (Course Link)

- GEOG/LDARCH C188: Geographic Information Systems (*Lead Instructor*) **Fall 2022**

- Teaching Effectiveness: 6.311/7 from 61/169 students (Dept. Average: 6.230/7)

Graduate Student Instructor

- LDARCH/ESPM C289: Applied Remote Sensing **Spring 2024**
- GEOG/LDARCH C188: Geographic Information Systems **Fall 2021**

Seoul National University

Teaching Assistant

- 457.542*: Advanced Surveying (*Head TA*) **Spring 2021**
- 457.205: Introduction to Geospatial Engineering (*Lab Tutor & Head TA*) **Spring 2021**
- 457.539*: Advanced Remote Sensing: VHR Imagery (*Head TA*) **Fall 2020**
- 457.402: Remote Sensing (*Lab Tutor & Head TA*) **Fall 2020**
- 457.544*: Satellite Image Interpretation (*Head TA*) **Spring 2020**
- Leadership for Civil Engineers (*TA*) **Spring 2020**
- 457.205: Spatial Informatics and Systems (*Lab Tutor & Head TA*) **Spring 2020**

**Graduate-level Courses*

Mentored Students at UC Berkeley

- Stella Wing (BS Conservation and Resource Studies & Data Science) **Sep 2023 – May 2024**
  Current: MESM @ UCSB
- Harrison Raine (MLA & MCP) **Sep 2023 – July 2024**
  Current: NASA
- Zeff Fengze Lin (Visiting student from South China University of Technology) **Jan 2023 – May 2023**
  Current: Ph.D. @ Tsinghua U
- Weixin Li (MS Civil & Environmental Engineering) **Sep 2022 – May 2023**
  Current: Ph.D. @ UCSB
- Xihan Yao (MLA) **Sep 2022 – May 2023**
  Current: Ph.D. @ UT Austin
- Madison Chi (BS Environmental Science & Sustainable Design) **Sep 2022 – May 2023**
  Current: MPH @ UCLA

Mentored Students at Seoul National University

- Hyoungwoo Choi (BS Civil and Environmental Engineering) **Sep 2020 – Feb 2021**

Services

Reviewer (Total: 49 reviews)

Agronomy (3), Applied sciences (2), European journal of remote sensing (1), Fire (2), Forecasting (3), Geospatial information science (2), Geocarto international (1), GIScience & remote sensing (3), IEEE JSTARS (2), International journal of Digital Earth (2), ISPRS international journal of geo-information (2), Land (2), Remote sensing (22), Sustainability (2)

Membership

- Association for Fire Ecology **2025**
- American Geophysical Union **2025**
- International Association of Wildland Fire **2025**
- Association of Environmental Professionals **2024**
- IEEE Geoscience and Remote Sensing Society **2023**
- American Society for Photogrammetry and Remote Sensing **2022**
- Korean-American Scientists and Engineers Association **2022**
- Korean Graduate Student Association **2021**
- International Society for Photogrammetry and Remote Sensing Student Consortium **2020**
- Korean Society of Civil Engineers **2019**

Session Chair (Image Analysis for Land Cover Mapping), IGARSS 2023 **July 2023**

Geospatial Data Consultant, Investigative Reporting Program **Sep 2022 – Dec 2022**
- Developed and managed multi-modal geospatial data (multispectral satellite images, nighttime light images, vector data related to census, parcels, etc.) to map deforestation and human activity in Brazil using Google Earth Engine (geemap & Javascript) and QGIS for J298 OSINT Seminar in the School of Journalism at UC Berkeley.
- Provided workshops and advised journalists on GIS and geospatial tools.

Ammunition Inspector, Republic of Korea Army **May 2017 – Jan 2019**
- Recorded ammunition transactions and composed ammunition inventory reports. After working hours, contributed to write-up on pan-sharpening image fusion research using Worldview images [P1].

General Education Peer Tutor, Seoul National University **Mar 2016 – June 2016**
- Tutored college-level English to undergraduate students for incoming freshmen

Section Editor, The SNU Quill - SNU's English Press **Sep 2013 – June 2015**
- SNU campus news section reporter and editor for 9 volumes; responsible for 6-8 journal reporters. Also coordinated English writing/composition workshops and orientations.

Projects

Roles: GSR/RA | Grant Writing | Project Manager

Awarded Projects

Multi-scale mitigation of wildfire risk vulnerabilities in the natural and built environments (College of Environmental Design, UC Berkeley)	\$35,000	**2025-2026**
Regional Sediment Bulking Methods for California in Support of Post Wildfire Flood Mitigation (California Department of Transportation)	\$400,000	**2022-2024**
Multiscale analysis for Improved Risk Assessment of Wildfires facilitated by Data and Computation (C3.ai)	\$200,000	**2021-2022**
Deep Learning Framework for Mapping Basic Spatial Data in North Korea Using Very High Resolution Satellite Imagery (Institute for Peace and Unification)	\$28,000	**2021-2022**
Development of Disaster Analysis Technology Using High-Resolution Satellite Imagery (Ministry of the Interior)	\$300,000	**2019-2022**
Forecasting Photovoltaic Power using Meteorological Satellite Imagery and Deep Learning (SK Telecom)	\$60,000	**2019**

Skills

GitHub	https://github.com/minhokim93
Coding	Python, C/C++, CMake, Matlab, Javascript (Google Earth Engine), Git, Scripting (Bash), LaTeX, HTML

ML/DL	Tensorflow, Keras, PyTorch, Scikit-learn, Scikit-image, OpenCV
Remote Sensing	ENVI (SARscape), Google Earth Engine (Javascript/geemap), Python (Rasterio/GDAL)
GIS	ArcGIS, QGIS, Python
Languages	English (Native), Korean (Native), French (Fluent)

Certification

- Applied Data Science **Expected**
- Teaching and Learning in Higher Education **Expected**
- Geospatial Information Science and Technology **2022**

References

Dr. Prof. Marta C. González
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University of California, Berkeley
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Dr. Prof. John Radke
Department of Landscape Architecture and Environmental Planning
Department of City and Regional Planning
University of California, Berkeley
412 Wurster Hall #2000
Email: ratt@berkeley.edu

Dr. Prof. Yongil Kim
Department of Civil and Environmental Engineering
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Avinguda de Serra Galliners, 112, 08290, Cerdanyola del Vallès, Spain
Email: incendi@yahoo.com